Tobruk University

Faculty Of Science

Final exam

Subject: waves and ultrasound

Time :2 hours

Name ..... Student No( )

Q1 .Choose the correct answer from the following :

1. Resonance occurs when the frequency of an external force is:
   A. Zero
   B. Higher than natural frequency
   C. Equal to the natural frequency
2. Superposition of waves occurs only when waves:
   A. Never meet
   B. Overlap in space and time
   C. Have zero amplitude
3. The Doppler effect occurs due to:
   A. Change in medium
   B. Reflection of sound
   C. Relative motion between source and observer
4. Beam formation is the process of:
   A. Increasing sound intensity in all directions
   B. Directing sound waves in a specific direction
   C. Absorbing sound completely
5. Absorption is the process in which a medium:
   A. Reflects wave energy completely
   B. Transmits wave without loss
   C. Takes in wave energy and converts it into other forms

Q2. How do they propagate Ultrasound in biological tissues?

Q3. Discuss the main components of an Ultrasound System.

Q4. What are the causes of attenuation in waves?

Q5. Discuss the Types of Reflection

Q6. A radio waves transmitted from a certain radio station is represented by the wave equation: $y = 0.75\sin(0.67\pi x - 2 \times 10^8\ \pi t)$ Calculate the (i) wavelength of the wave (ii) frequency of the wave (ii) velocity of the wave.

$$y = 0.75\sin(0.67\pi x - 2 \times 10^8\ \pi t)$$

Q7. A train whistle produces a sound with a real frequency of (f = 600 HZ)
The frequency heard by the observer is 652.9 HZ ,
The source is stationary. The speed of sound in air is (c = 340 , m/s)

Calculate the speed of the observer

Good luck

Tobruk University

Faculty Of Science

Final exam

Subject: Waves and ultrasound Time:2 hours

Q1 Choose the correct answer from the following :

1. Which statement correctly describes ultrasound waves?

A. They are electromagnetic waves that can travel through a vacuum.

B. They are mechanical longitudinal waves with frequencies greater than 20 kHz and require a material medium to propagate.

C. They are transverse waves with frequencies below 20 kHz.

2.What is the piezoelectric effect?

A. The conversion of heat energy into electrical energy.

B. The conversion of mechanical and electrical energy into each other in certain crystalline materials.

C. The conversion of light energy into sound energy.

3.What is a standing wave?

A. A wave formed by the interference of two waves with the same frequency and amplitude traveling in opposite directions.

B. A wave that travels in one direction only.

C. A wave that can only travel through a vacuum.

4.What is beam formation in ultrasound?

A. The process of directing sound waves in a specific direction instead of allowing them to spread in all directions.

B. The process of increasing the frequency of sound waves.

C. The process of converting sound waves into light waves.

5.What is the Doppler effect?

A. The change in frequency (or wavelength) of a sound wave due to relative motion between the source and the observer.

B. The reflection of sound waves from a solid surface.

C. The absorption of sound waves by body tissues.

Q2: Explain the working principles and characteristics of both Continuous Doppler and Pulsed Doppler.

Q3: An ambulance produces a sound with a real frequency of (f = 500 , Hz). The ambulance is moving towards a stationary observer at a speed of (v source = 20 , m/s). The speed of sound in air is (c = 340 , m/s).

Calculate the frequency heard by the observer.

Q4: How do they propagate Ultrasound in biological tissues?

Q5: Derive the equation of a standing wave formed by the superposition of two harmonic waves of equal amplitude and frequency traveling in opposite directions.

Q6: What are the conditions required for the principle of superposition to occur? Explain briefly.

Q7: What are the main causes of wave attenuation?

Tobruk University

Faculty Of Science

Date: 7/5/2026 Partial exam

Subject: wave and ultrasound Time : 1:30

Name ..... Student No( )

Q1. For each of the following definitions, provide the appropriate term :

1. is a disturbance which travels through a medium transferring energy from one point to another without causing any permanent displacement of the medium.
2. is a wave in which travel perpendicularly to the direction of the vibrations producing the waves.
3. is the bouncing back of a sound wave into the original medium when it strikes a surface or boundary.
4. is a physical phenomenon that occurs when a system is driven by an external periodic force whose frequency matches the system's natural frequency.
5. the ratio of acoustic pressure to particle velocity in a medium.

Q2. Choose the correct answer from the following :

1. Which of the following factors does the speed of sound in liquids depend on (a) Color and temperature (b) Density and compressibility (c) Volume and shape
2. What is an echo (a) A sound that is absorbed by walls (b) A distinct repetition of sound due to reflection (c) A decrease in sound intensity over distance
3. which of the following is a factor affecting sound reflection?(a) Color of the object (b) Nature of the surface (hard or soft)(c) Shape of the sound source
4. A standing wave is formed when two waves:(a) have different amplitudes and same direction (b) have the same frequency and amplitudes travel in opposite directions (c)Have different frequencies and same direction

5. What is a longitudinal wave (a) A wave that travels perpendicular to the vibrations of the medium (b) A wave that travels parallel to the vibrations of the medium (c) A wave that does not require a medium to travel.

Q3. Explain the main criteria used in classifying waves?

Q4. Derive the equation of a standing wave?

Q5. What Happens During Transmission?

Q6. A radio waves transmitted from a certain radio station is represented by the wave equation: $y = 0.75\sin(0.67\pi x - 2 \times 10^8\ \pi t)$ Calculate the (i) wavelength of the wave (ii) frequency of the wave (ii) velocity of the wave.